

Shreyas Nalle

portfoliowebsite/ShreyasNalle | x/ShreyasNalle | github/Shreyasnalle | linkedin/Shreyas-nalle | shreyas.nalle7@gmail.com

TECHNICAL SKILLS

Languages : JavaScript, TypeScript, Python, C, C++, Java, Golang

Frontend : Next.js, React, Tailwind CSS, shadcn/ui, Motion, GSAP

Backend and Frameworks : FastAPI, Node.js, Express.js, Bun, Pydantic

AI & Machine Learning : LangChain, LangGraph, PyTorch, scikit-learn

Databases : PostgreSQL, MySQL, Redis, pgvector, MongoDB, Prisma

Developer Tools & Platforms: Supabase, Figma, Vercel, Render, Stripe, Docker, Vite, Git, Linux, Resend, Jupyter

PROJECTS

[Simply – In tab RAG Assistant](#) | FastAPI, JavaScript, Chrome Extension, pgvector, PostgreSQL

- A Chrome extension intercepting live YouTube captions, routing them through a RAG pipeline and storing conversation in relational databases.
- Shipped the extension with version v1.0.1 to Microsoft Edge Add ons store.
- Currently working on integrating more features and an agentic RAG system.

[Ramplimg – Backend Production Readiness Reviewer](#) | Next.js, FastAPI, Go, Semgrep, k6, PostgreSQL, Redis, Docker

- An automated code analysis that runs on any repo, builds AST dependency graphs and executes context-aware Semgrep rules inside sandboxed Docker containers.
- Autonomous agents execute k6 load-tests across staging environments, depending on their results, which reloops into Semgrep run with new rules, for better insights.
- Synthesized dynamic load metrics and static diagnostics form up a report, with attached md file, thus leveraging AI to map concurrency thresholds and exact line-level bottlenecks to work on.

[GenAI Fraud Defense Autonomous Red/Blue Team](#) | PyTorch Geometric, Heterogeneous GNNs, FastAPI, Next.js, Groq API, Docker

- An adversarial training loop to identify financial fraud patterns by LLM-driven Red Team attack generator with an adaptive Blue Team Heterogeneous Graph Neural Network (Multi-GNN) model.
- Scaled graph representation pipelines across 5M+ transaction edges and 1.75M + account nodes, training PyTorch Geometric GINe layers across 14+ currency and 6+ payment methods.
- Defenses against 5 GenAI amplified attack vector types, automating false negative mining and closed-loop model fine-tuning with class weighted cross entropy loss.

OPEN SOURCE CONTRIBUTION

OpenSRE (Tracer-Cloud)

- Resolved message truncation and rendering failures in the Telegram alerting gateway by refactoring downstream Markdown-to-HTML conversion logic to adhere to API character payload limits.
- Enhanced production observability delivery across multi-channel alerting integrations, ensuring reliable formatting and zero silent text fallback during live incident streaming.

ACHIEVEMENTS

1. 4th Place at Hack-it-Out Hackathon DPU Pimpri (Zion 2026)
2. Special Mention at DPU MUN 2025

3. TENET MUN 2025 Marketing head